

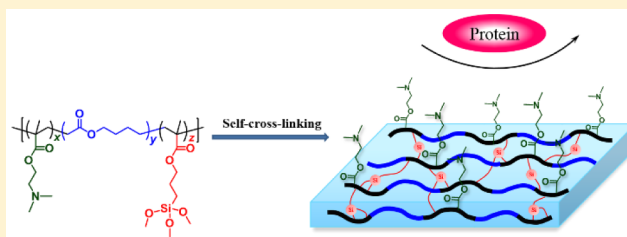
Self-Cross-Linking Degradable Polymers for Antifouling Coatings

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 Supporting Information

ABSTRACT: Degradable polymers with protein resistance can find applications in antibiofouling. We have prepared copolymer of 2-methylene-1,3-dioxepane (MDO), 2-(dimethylamino) ethyl methacrylate (DEM) and 3-(methacryloxypropyl) trimethoxysilane (KH570) via radical ring-opening polymerization, where MDO, DEM, and KH570 make the polymer degradable, protein resistant and self-cross-linkable, respectively. Our studies demonstrate that the self-cross-linking significantly improves the coating ability of the polymer with controlled biodegradation in seawater. We have investigated the adsorption of fibrinogen, bovine serum albumin and lysozyme on the self-cross-linking polymer surface as a function of its composition by use of quartz crystal microbalance with dissipation (QCM-D). It shows the polymer network can resist the adsorption of proteins in seawater. The antibacterial adhesion of the polymer network was evaluated by using *Micrococcus luteus* (*M. luteus*) and *Pseudomonas* sp., revealing that the polymer network can effectively inhibit the settlement of marine bacteria.



■ INTRODUCTION

Nonspecific protein adsorption is a critical issue in biomedical devices,¹ water purification,² food packaging,³ and textiles.⁴ Protein adsorption also brings a great challenge to marine industries since marine biofouling starts with formation of biofilm consisting of proteins and glycoproteins.^{5,6} Accordingly, protein resistant materials such as polysaccharides,⁷ poly(2-oxazoline),^{8,9} poly(ethylene glycol) (PEG),^{10–12} and zwitterionic polymers^{13,14} have been developed in the past years. Generally, they are hydrophilic polymer with zero net charge, and their nonfouling capability depends on the hydration layer which can effectively reduce the hydrophobic interactions between the proteins and polymer surface.^{15,16} Particularly, poly[2-(dimethylamino) ethyl methacrylate] (PDEM) exhibits excellent protein resistance and antibacterial capability in marine environment due to its hydrophilic nature and fully deprotonated state.¹⁷ However, since it cannot degrade in marine environment,¹⁸ unwanted inorganic fouling accumulation or the deposition of dead bacteria would change the protein resistant surface, leading to the subsequent formation of biofilm and development of biofouling.¹⁹ The nondegradable polymers can also cause ecology problems. Recently, we have developed marine antibiofouling materials based on biodegradable polymers, where the degradation of polymer in marine environment leads to a dynamic surface that can prevent the settlement of marine biofouling.^{20–24} Therefore, degradable PDEM based materials which can form dynamic surface are desired.

Copolymerizing cyclic ketene acetals (CKAs) with vinyl monomers via radical ring-opening polymerization provides a simple and versatile way to generate a new class of degradable poly(vinyl-co-esters).²⁵ As one member of CKAs family, 2-methylene-1,3-dioxepane (MDO) was copolymerized with

glycidyl methacrylate (GMA),^{26,27} oligo(ethylene glycol) methacrylate (OEGMA),²⁸ N-isopropylacrylamide (NIPAM).²⁹ It is a useful platform to develop functional degradable poly(ϵ -caprolactone)-based material for drug carrier, medical adhesives and scaffold for tissue engineering.³⁰ However, such poly(vinyl-co-esters) copolymers with low crystallinity and low glass transition temperatures (T_g) often exhibit weak bulk mechanical properties and poor coating ability.

In the present study, we have prepared degradable copolymer consisting of MDO, 2-(dimethylamino) ethyl methacrylate (DEM) and 3-(methacryloxypropyl) trimethoxysilane (KH570) by radical ring-opening polymerization. The presence of MDO and DEM makes the polymer degradable and protein resistant, whereas KH570 yields self-cross-linking via heating or moisture.^{31,32} The polymer forms three-dimensional network after the post self-cross-linking so that its mechanical strength and coating ability are improved. Such degradable network with protein resistance yielding dynamic surface can effectively inhibit the adsorption of microorganisms in marine environment without addition of antifoulants. We have investigated the biodegradability and antifouling performance of the polymers. Our aim is to develop environment-friendly systems for antibiofouling.

■ EXPERIMENTAL SECTION

Materials. 2-(Dimethylamino) ethyl methacrylate (DEM) from Aladdin was reflux over CaH_2 and distilled prior to use. 2-Methylene-1,3-dioxepane (MDO) was synthesized by previous

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Scheme 1. Radical Ring-Opening Polymerization of MDO, DEM and KH570

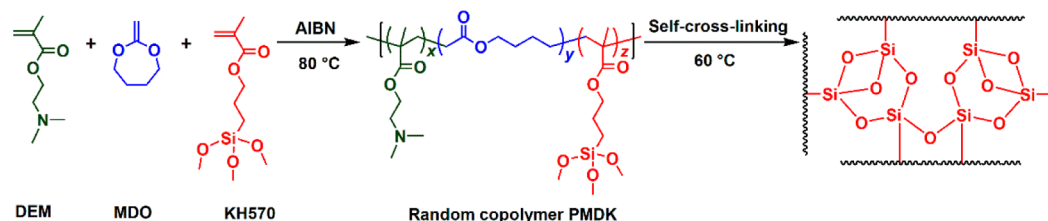


Table 1. Characterization Data of PMDKx

sample	MDO: DEM: Si ^a	MDO: DEM: Si ^b	M _n (kDa) ^c	Đ _M ^c	T _g (°C) ^d	γ(MPa) ^e
PMDK16	70:13:17	16:73:11	12.1	2.1	6.2	1.35 ± 0.21
PMDK47	82:14:4	47:42:11	6.9	2.2	−23.7	1.05 ± 0.13
PMDK64	90:7: 3	64:26:10	5.5	2.5		0.76 ± 0.09
PMDO	100:0: 0	100:0: 0	14.0	2.8		

^aFeed molar ratio. ^bMolar ratio determined by ¹H NMR. ^cDetermined by GPC. ^dDetermined by DSC. ^eAdhesion strength

procedure described elsewhere.³³ 3-(Methacryloxypropyl) trimethoxysilane (KH570) from Aldrich was used as received. AIBN (Aladdin) was recrystallized from methanol twice. 1,4-Dioxane was dried by CaH₂ and distilled prior to use. Artificial seawater (ASW) was prepared following ASTM D1141–98 (2013). Lipase from *Pseudomonas cepacia* (PS) was from Aldrich. Fibrinogen (fraction I from human plasma, M_w = 340 kDa, pI 5.5), lysozyme via chicken egg white (M_w = 14.7 kDa, pI 11.1), and bovine serum albumin (fraction V from the crystalline form, M_w = 68 kDa, pI 4.8) were purchased from Merck Chemicals.³⁴ Scheme 1 shows the synthesis of copolymer of MDO, DEM and KH570.

Synthesis of Copolymer of MDO, DEM and KH570. As described in Scheme 1, the polymers were synthesized in the solution of 1,4-dioxane. Briefly, a certain amount of MDO, DEM, KH570, and AIBN (1 mol % of the total monomers) were dissolved in 1,4-dioxane and added into a Schlenk tube, then the mixture was degassed by three vacuum/nitrogen cycles. The polymerization was performed under nitrogen at 80 °C for 24 h. The solution was condensed and precipitated in hexane. After drying for 24 h, the polymers are obtained. ¹H NMR (600 MHz, CDCl₃, ppm): 3.88–4.22 (–COOCH₂–), 3.58 (–Si(OCH₃)₃), 2.57 (–N(CH₃)₂CH₂–), 0.89 (–COOCH₂CH₂CH₂CH₃), 0.65 (–CH₂Si(OCH₃)₃). The copolymer is designed as PMDK_x, where *x* is the molar percentage of MDO determined by calculating the integral ratio of –CH₂COOCH₂– of MDO units (δ = 2.30), (–N(CH₃)₂CH₂–) of DEM units (δ = 2.57) and –Si(OCH₃)₃ of KH570 units (δ = 3.58), the details can be found in Supporting Information (Figure S2). The characterization data of PMDK_x are shown in Table 1. For comparison, the PMDO homopolymer was also prepared. The product is viscous paste after purification. ¹H NMR(600 MHz, CDCl₃, ppm): 4.02 (–COOCH₂–), 3.26(–CH₂COO–), 1.60(–COOCH₂CH₂CH₂CH₂–), 1.34 (–COOCH₂CH₂CH₂CH₂–), 0.88 (–COOCH₂CH₂CH₂CH₃).

■ CHARACTERIZATION

Proton Nuclear Magnetic Resonance Spectroscopy (¹H NMR). All ¹H NMR spectra were recorded on a Bruker AV600 NMR spectrometer using CDCl₃ as solvent and tetramethylsilane (TMS) as internal standard.

Fourier Transform Infrared Spectroscopy (FTIR). FTIR spectra were recorded on a Bruker Vertex 70 FTIR

spectrometer. The spectra were collected at 64 scans with a spectral resolution of 4 cm^{−1} by KBr disk method.

Thermal Analysis. The Differential scanning calorimetry (DSC) measurement was performed on a NETZSCH DSC 204F1 differential scanning calorimeter under a nitrogen flow of 50 mL/min. Each sample was quickly heated to 120 °C and equilibrated at the temperature for 5 min to remove thermal history, then cooled to −80 °C and kept for 15 min. Then, it was heated to 110 °C at a rate of 10 °C/min. All DSC traces were from the second heating process.

Gel Permeation Chromatography (GPC). The number-average molecular weights (M_n) and the polydispersity (Đ_M) were determined by GPC coupled with successively connected UV and RI detectors at 35 °C using two identical PLgel columns (5 μm, MIXED-C). Tetrahydrofuran (THF) was used as the eluent at the flow rate of 1.0 mL/min. Calibrations were performed using polystyrene standards.

Quartz Crystal Microbalance with Dissipation (QCM-D). The adsorption of protein was measured by QCM-D from Q-sense AB (Sweden). PMDK_x film was prepared by spin-casting from polymer solution in THF (5.0 mg/mL) on AT-cut quartz crystal surface, and then cured at 60 °C for 48 h. The quartz crystal was placed in a fluid cell with the side of polymer film exposed to the solution. The protein solution (1.0 mg/mL) was added to the surface at a flow rate of 150 μL/min using ASW as the baseline. The details about QCM-D measurements are described elsewhere.³⁵ Briefly, the shift in frequency (Δ*f*) of the crystal is associated with the mass change of the thin layer on quartz crystal, whereas the shift in dissipation (Δ*D*) is related to the viscoelastic properties of the additional layer.

For a rigid film in vacuum or air, if it is evenly distributed and much thinner than the crystal, change in frequency (Δ*f*) is related to Δ*m* and the overtone number (*n* = 1, 3, 5...) by the Sauerbrey equation,³⁶

$$\Delta m = -\frac{\rho_q l_q \Delta f}{f_0 n} = -C \frac{\Delta f}{n}$$

Where *f*₀ is the fundamental frequency, ρ_q and *l*_q are the specific density and thickness of the quartz crystal, respectively, and *C* is the constant of the crystal (17.7 ng/cm²·Hz). Here, they give information about the protein adsorption and structural change of PMDK_x films. All the experiments were

performed at 25 °C and the data were from the third overtone ($n = 3$).

Antibacterial Assays. Marine Gram-positive *Micrococcus luteus* (*M. luteus*) and Gram-negative *Pseudomonas* sp. were used to evaluate the antibacterial ability of the polymer coatings. Bacterial suspension was prepared with ASW at a concentration of 10^8 cells/mL. Each substrate with PMDKx film was then incubated in 1 mL of bacterial suspension under static condition at 30 °C for 9 h. The surface of PMDKx film after immersion in bacterial suspension was stained with LIVE/DEAD BacLight Bacterial Viability Kit. The adhered bacterial cells were examined by a fluorescence microscope (Scope A1, Zeiss).

Enzymatic Degradation. The tests were conducted in 0.1 mg/mL Lipase from *Pseudomonas cepacia* (PS) solution of ASW. PMDKx films were prepared by casting from xylene on an epoxy panel ($20 \times 20 \text{ mm}^2$) and cured at 60 °C for 48 h. The weight (W_0) of each dried coating together with its panel was measured before the coating was immersed in PS solution which was replaced every 3 days. At certain intervals, the sample was taken out, dipped in Milli-Q water for 5 min, then dried before their weights ($W_{t,\text{dry}}$) were recorded. The mass loss (wt %) was calculated by the following equation:

$$\text{mass loss (wt\%)} = \frac{W_0 - W_{t,\text{dry}}}{W_0 - W_{\text{panel}}} \times 100\%$$

where W_0 and $W_{t,\text{dry}}$ are the initial weight and dry weight at time t of the coated panel, respectively. Each sample was measured in triplicates.

Adhesion Test. The adhesion strength of PMDKx coatings to the glass fiber reinforced epoxy resin substrate was measured by adhesion tester (PosiTest AT-A automatic) according to ASTM Standard D 4541. PMDKx films with a thickness of $\sim 300 \mu\text{m}$ were prepared by casting from polymer solution and then cured at 60 °C for 48 h. Pull-test adhesion data was obtained by detaching an aluminum dolly from testing area of 20 mm in diameters at speed of 0.2 MPa s^{-1} . Five different regions for each sample were tested to obtain an average value.

■ RESULT AND DISCUSSION

The successful synthesis of PMDO homopolymer and PMDKx copolymers was confirmed by ^1H NMR (Supporting Information Figures S1 and S2). For the samples before cross-linking, we can observe a peak at 0.89 ppm. It is attributed to alkyl branches occurring in intramolecular hydrogen transfers during radical propagation.^{37,38} Such short branches can disturb chain regularity, resulting in less crystalline or even totally amorphous structure compared to the semicrystalline poly(ϵ -caprolactone), which can facilitate faster degradation.^{39,40} Yet, it would reduce the mechanical properties and coating ability of the polymer. On the other hand, as MDO content increases, the T_g decreases (Table 1), PMDKx even exhibits T_g below room temperature (-23.7 and 6.2 °C for PMDK47 and PMDK16, respectively). Such PMDKx is viscous paste and can not be used as coating without cross-linking.

Figure 1 shows FTIR spectra of PMDK16 before and after self-cross-linking via heating with moisture. The band at 823 and 1089 cm^{-1} due to the Si-O-CH_3 in PMDK16 disappear and the new one at 1107 cm^{-1} assigned to the $-\text{Si-O-Si}-$ in polymer network appears, indicating the complete cross-linking of the polymer via KH570 units.⁴¹ Correspondingly, PMDKx changes from viscous paste to solid film after self-cross-linking.

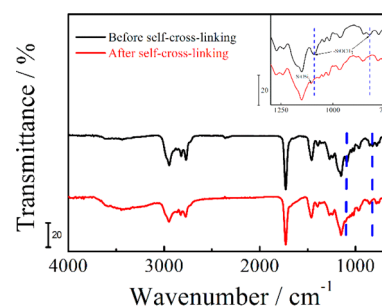


Figure 1. FTIR spectra of PMDK16 before and after heating at 60 °C for 48 h. The insert graph is the magnified spectra in the wavenumber region from 1300 to 700 cm^{-1} .

The corresponding data of adhesion strength are shown in Table 1. The adhesion strength increases with DEM content. This is because DEM are polar groups and can enhance the interfacial interactions between substrate and coatings.⁴²

The degradation of polymer is important for antifouling performance since its self-renewal property can polish away the adsorbed fouling organisms. Using biodegradable polymers is also favorable to the marine environment. Actually, there are various microorganisms secreting enzyme in natural marine environment, making the biodegradation of ester-containing polymers possible.⁴³ We examined the enzymatic degradation behaviors of PMDKx in the presence of Lipase PS (Figure 2).

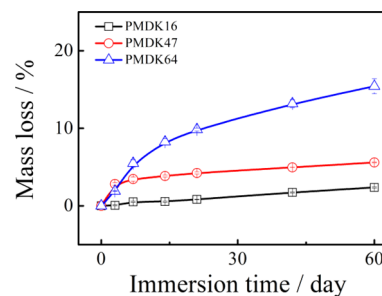


Figure 2. Time dependence of mass loss for PMDKx in ASW with PS lipase (0.1 mg/mL) at 25 °C.

After immersion in ASW for 3 days, all films start to lose weight due to cleavage of ester linkage in the main chain and consequent dissolution of small fragments. As the MDO content increases, the mass loss increases. It is known that the degradation rate can be affected by the content of liable linkage, wettability, cross-link density and so on.⁴⁴ In this study, since all the PMDKx samples have similar cross-linking density (Table 1, Figure 1 and Supporting Information Figure S3), the biodegradation is determined by MDO content. Anyhow, all the PMDKx films can degrade in ASW and the biodegradation rate of films can be readily adjusted by MDO content. Note that the time dependence of mass loss for PMDO is not given, this is because PMDO is viscous paste with weak adhesion to substrate and detached from the substrate after 7 days in ASW.

The protein adsorption on PMDKx surface was evaluated by QCM-D. Figure 3 shows the time dependences of frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on PMDKx surfaces at 25 °C. Fibrinogen is a large plasma protein carrying negative charge in ASW (pH 8.2), which can strongly adsorb onto hydrophobic surfaces.³⁴ Note that the charges of proteins or lipase decreases as the environment changes from physiological conditions to ASW

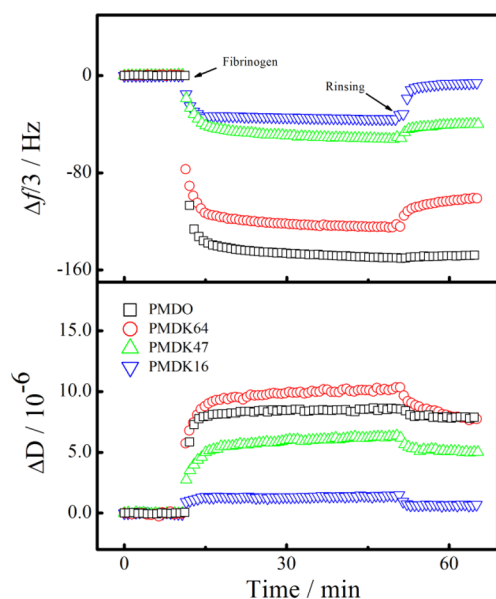


Figure 3. Frequency shift (Δf) and dissipation shift (ΔD) for adsorption of fibrinogen on PMDKx films.

with higher pH and much higher salinity, which leads protein adsorption to increase due to the reduction of electrostatic repulsion. On the other hand, the hydration of proteins or lipase would decrease with ionic strength, leading structure to be more compact and the activity to decrease.^{45–47} However, to provide valuable data for marine antibiofouling, the experiments must be performed in ASW. This does not affect the interpretation of the results because the effect of pH and salinity on the proteins or lipase are the same for each PMDKx sample. For PMDO, after fibrinogen is introduced, Δf decreases sharply and then gradually levels off. After rinsing with ASW, Δf still exhibits a marked decrease in comparison with the baseline for ASW. It is known that the frequency (Δf) decreases with the mass or thickness of the layer, whereas the dissipation (ΔD) increases with the thickness but decreases with the rigidity of the layer.⁴⁸ The marked decrease in Δf (450 Hz) indicates the considerable adsorption of fibrinogen on the PMDO surface. According to Sauerbrey equation,³⁶ the mass of fibrinogen adsorption is $\sim 2655 \text{ ng/cm}^2$. Besides, the notable increase in ΔD when fibrinogen is introduced further indicates that fibrinogen is adsorbed and forms a viscoelastic layer. As DEM content in PMDKx increases, the changes in Δf and ΔD after rinsing become smaller compared to those for PMDO surface, indicating the protein resistance of PMDKx to fibrinogen increases. When DEM content increases up to 73% (PMDK16), the decrease in Δf is 15 Hz after rinsing and the corresponding mass of protein adsorption is $\sim 88.5 \text{ ng/cm}^2$, which indicates that PMDK16 can reduce the fibrinogen adsorption over 97% compared to PMDO film. Clearly, the protein resistance increases with DEM content. This is because DEM groups is hydrophilic and fully deprotonated in ASW,¹⁷ which can reduce hydrophobic interactions between fibrinogen and polymer surface. As reported before, zwitterionic polymers by betainization of DEM groups exhibit excellent protein resistance. However, it is difficult to introduce highly polar zwitterions to the hydrophobic PCL backbone since they are not compatible and they do not have the same solvent.^{49,50} In seawater, the DEM groups exhibit similar protein resistance

with the zwitterionic sulfobetaine groups, but have better compatibility with hydrophobic PCL segment.

Figure 4 shows the changes in Δf and ΔD for the adsorption of BSA on PMDKx surfaces. BSA is a smaller protein with

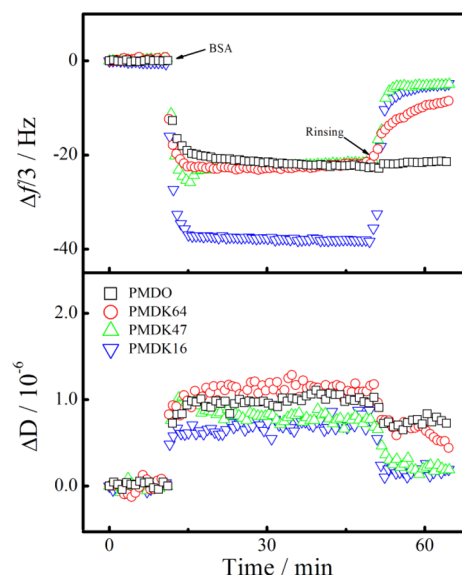


Figure 4. Frequency shift (Δf) and dissipation shift (ΔD) for adsorption of bovine serum albumin (BSA) on PMDKx films.

negative charge in ASW.⁴⁷ For PMDO, the marked decrease in Δf is 63 Hz ($\sim 317.7 \text{ ng/cm}^2$) relative to the baseline indicates that BSA is adsorbed. As DEM content increases, the changes in Δf and ΔD become smaller. When DEM content is above 42%, the decrease in Δf is $\sim 9 \text{ Hz}$, indicating that PMDK16 can reduce the BSA adsorption over 85% compared to PMDO film.

Figure 5 shows the changes of Δf and ΔD for the adsorption of lysozyme. Lysozyme is the smallest protein among the three with positive charges in ASW.⁵¹ Like fibrinogen and BSA, lysozyme is absorbed onto PMDO surface, reflected by the marked decrease in Δf and increase in ΔD . As DEM content

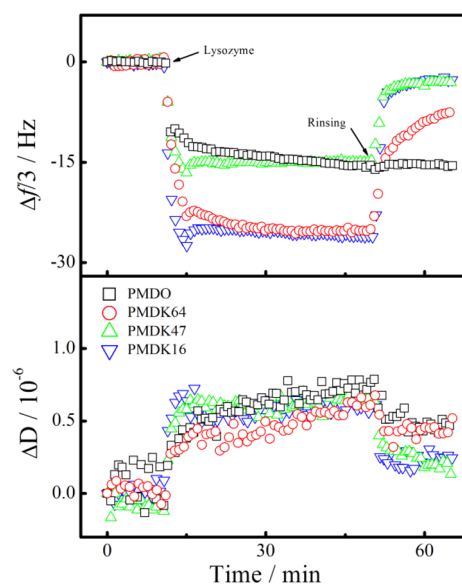


Figure 5. Frequency shift (Δf) and dissipation shift (ΔD) for adsorption of lysozyme on PMDKx films.

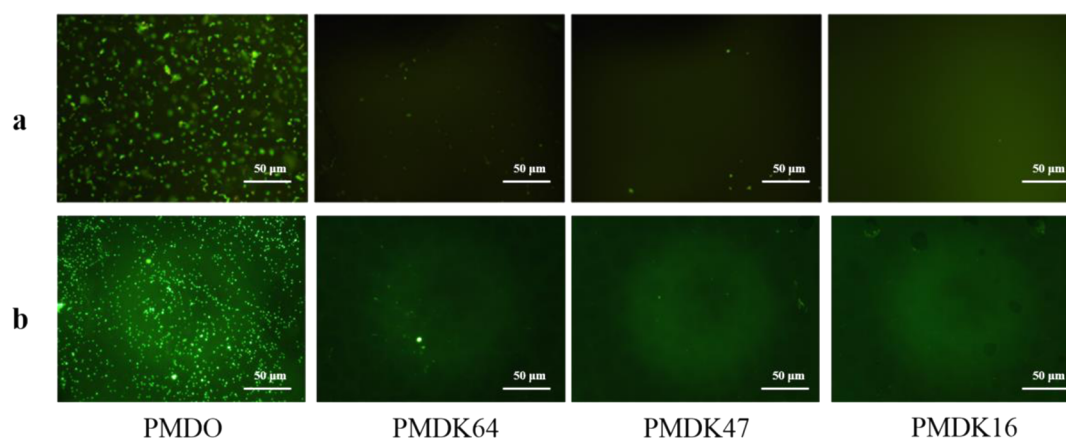


Figure 6. Fluorescence microscopy images taken from self-cross-linking PMDKx surfaces immersed in (a) *M. luteus*, (b) *Pseudomonas* sp. suspension for 9 h.

increases, the changes in Δf and ΔD after rinsing decreases, indicating the increased protein resistance to lysozyme. When DEM content is above 42%, the mass of lysozyme adsorption decreases from 265.5 to 35.4 ng/cm². It was reported that DEM-containing polymer can resist the adsorption of lysozyme due to the electrostatic repulsion between the positively charged lysozyme and positively charged PDEM.⁵⁰ However, the DEM groups in ASW carry zero net charge, so the protein resistance in this study arises from the hydrophilicity and full deprotonation of PDEM in ASW.¹⁷

Figure 6 shows the fluorescence microscopy images for antibacterial adhesion assay, where the bacteria are marine Gram-positive *Micrococcus luteus* (*M. luteus*) and Gram-negative *Pseudomonas* sp.. For PMDO, a large quantity of bacterial cells is adsorbed to polymer surface, indicating that the polymer is highly susceptible to bacterial adhesion and colonization. For PMDKx, the number of attached bacteria significantly decreases. Clearly, the introduction of DEM moieties can improve the antibacterial ability. As discussed above, the antibacterial adhesion of PMDKx is associated with their protein resistance. This is probably because the bacterial adhesion generally begins with the adsorption of extracellular polymeric substances (EPS) mainly consisting of proteins and glycoproteins.^{52,53} A protein resistant surface can reduce the initial attachment and delay the colonization of microbes.^{54,55}

CONCLUSIONS

We have prepared copolymer of 2-methylene-1,3-dioxepane (MDO), 2-(dimethylamino) ethyl methacrylate (DEM) and 3-(methacryloxypropyl) trimethoxysilane (KH570) via radical ring-opening polymerization. The polymer is degradable, protein resistant and self-cross-linkable. The self-cross-linking allows the polymer to form network with enough coating ability. The cured polymer network shows controllable enzymatic degradation depending on the composition. The polymer exhibits excellent protein resistance and antibacterial adhesion in seawater.

ASSOCIATED CONTENT

Supporting Information

This material is available free of charge via Internet at The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.7b00557.

¹H NMR spectra, FTIR spectra of PMDK47 and PMDK64 before and after cross-linking (PDF)

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Notes

The authors declare no competing financial interest.

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